Foundations of ADCS

0. Course Introduction

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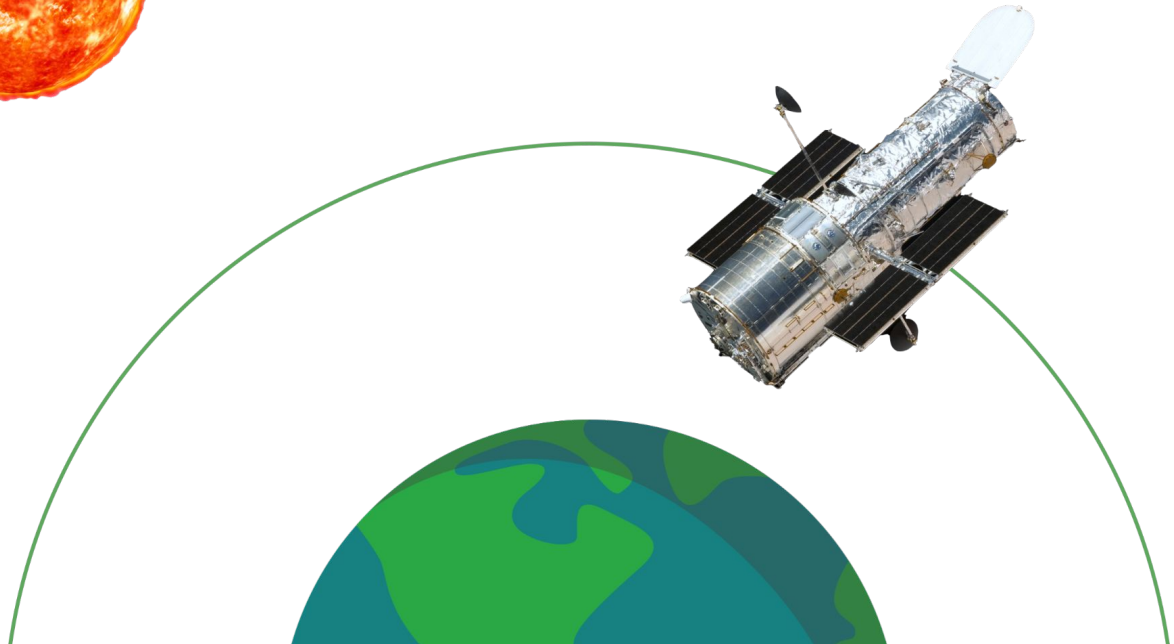
What is ADCS?

Altitude ≠ Attitude = Orientation



What is ADCS?

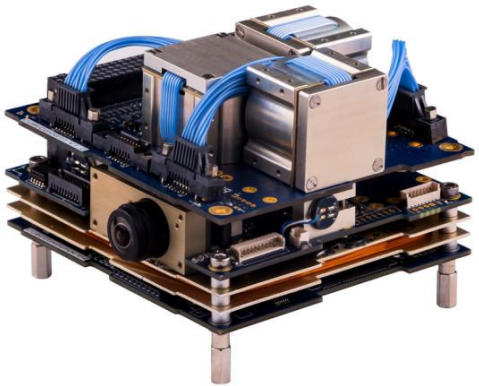
Attitude Determination
finding the orientation of spacecraft

Attitude Control
steering the spacecraft to desired orientation

Why ADCS?

*ADCS exploits the **laws of physics** using **mathematical algorithms** and involves the interaction of the **embedded hardware** with the **environment of space**.*

1. Applied mathematics and physics
2. Environment modelling
3. Control systems and estimation
4. Numerical simulations
5. Embedded hardware
6. Embedded firmware
7. Mechanical engineering
8. Material engineering
9. . . .



CubeADCS 3-Axis by CubeSpace

Challenges

1. Steep learning curve (not difficult but requires many preliminaries)
2. Highly technical resources and rare accessible expositions
3. Less popular and scarce open-source projects
4. Most of the implementations are proprietary to companies
5. Hard to draw line between theory and implementation

Opportunities

(Determination)

	Dynamics	Estimation
Orbit	Newton (1642–1727) Lagrange (1736–1813) Hamilton (1805–1865) Einstein (1874–1955)	Kepler (1571–1630) Lagrange (1736–1813) Gauss (1777–1855)
Attitude	Euler (1707–1783) Cayley (1821–1895)	(gone fishin')

… there is no great hoary founder of attitude estimation. My field would have to depend on rather more puny personages for its advancement, people like ***you*** and me.

Malcolm Shuster - Beyond Estimation (2005)

Opportunities

Knowledge of ADCS opens many opportunities

1. ADCS is rich in beautiful unsolved problems for researchers
2. Possibility of startups on ADCS products and services
3. Blog posts, articles, podcasts, and accessible expositions of ADCS
4. Contributions to opensource projects
5. Fewer competitions for job positions in ADCS.

Goals of this series

Popularize the field of ADCS in Nepal and beyond

1. Introduction to the preliminaries and foundations of ADCS
2. Logical progression to the problem statements of *attitude determination* and *attitude control system* respectively
3. Discussions of the state-of-art algorithms in ADCS
4. Mathematical models implemented on-board the satellites for ADCS
5. Basic simulation of the attitude control system
6. Appreciate how ADCS works in actual hardware and draw line between theory and implementation
7. Share references for further studies and research

Syllabus

https://github.com/risherlock/ADCS

0. Course introduction
1. Preliminaries of ADCS
2. Attitude determination system
3. Onboard models
4. Attitude control system
5. ADCS simulation
6. Discussions and conclusion

2 hours for each topic

Thank you!